

Aritra Roy | CV

Department of Mechanical Engineering, IIT Bombay

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Education

Indian Institute of Technology (IIT) Bombay, Mumbai, India 2025 – 2027 (Ongoing)
M.Tech. in Mechanical Engineering (*Thermal and Fluids Engineering*) CPI: **9.35/10**

Indian Institute of Technology (Indian School of Mines) Dhanbad, India 2020 – 2025
B.Tech. in Mechanical & Mining Machinery Engineering (Double Major) CPI: **9.08/10**[*]

[*]Weighted average of First Major (325 credits, 8.99 OGPA) and Second Major (152 credits, 9.28 OGPA)

Publications

A. Roy[*], A. Poddar, "Enhancing Electrokinetic Energy Harvesting via Rheology and Slip-Responsive Zeta Potential", *Proceedings of the Royal Society A*, vol. 482, 2026.

A. Roy[*], A. Mukherjee, B. Prasad, A. K. Nayak, "A computational analysis of flow dynamics and heat transfer in a wavy patterned channel using physics-informed neural networks", *Physics of Fluids*, vol. 37, issue 4, 2025.

A. Roy[*], S. Sengupta, A. K. Samanta, P. V. S. S. Likhith, S. K. Das, "Prospects of energy-efficient power generation system with ammonia as hydrogen carrier", *International Journal of Hydrogen Energy*, vol. 71, pp. 131-142, 2024.

Preprints & Manuscripts Under Review

A. Roy[*], A. Poddar, R. Valani, "Active particle dynamics in uniform and patterned electroosmotic flows", *under review in Journal of Fluid Mechanics*, arXiv:2611.00600 [physics.flu-dyn], 2026.

A. Roy[*], D. Das, P. Mondal, "Correlative effects of induced magnetic field-buoyancy on reactive solute dispersion dynamics in couple-stress fluids", *under review in Physical Review Fluids*, arXiv:2609.11099 [physics.flu-dyn], 2026.

Academic Honors and Fellowships

- Awarded the ***S N Bhatt Memorial Excellence Fellowship*** by ICTS-Tata Institute of Fundamental Research for exceptional performance in research related to geophysical turbulence. (Jul '24)
- Awarded the ***Chanakya UG Fellowship*** by AWaDH, IIT Ropar (Department of Ssience and Technology, India) for technology innovation in Cyber-Physical Systems. (Oct '22)
- Awarded Best Thesis Presentation (*3-Minute Thesis*) at the ***Research & Industrial Conclave '22***, IIT Guwahati, for inventing a bio-magnetically navigable needle for internal carotid artery haemodynamics. (Oct '22)

Work & Research Experience

- **Research Intern (Remote)** | *Rudolf Peierls Centre for Theoretical Physics, Oxford, UK* (May '25 – Oct '26)
  Guide – Dr. Rahil Valani, Department of Physics
  - Modeling active particle behavior with shape dependent interaction effects in electrohydrodynamically induced vortex flows under varying zeta potential.
- **Research Associate** | *International Center for Theoretical Sciences - Tata Institute for Fundamental Research (ICTS-TIFR), Bangalore,India* (May '24 – Ongoing)
  S N Bhatt Fellow | Guide – Prof. Rama Govindarajan, Fluid Dynamics and Turbulence division
  - Formulated and built an inhouse Method of Fundamental Solutions (MFS) solver for Oseen flow, for inertial sedimentation of dibilateral objects possessing dual planes of symmetry.
- **Research Project** | *Guide – Prof. Saikat Mukherjee (prior Postdoctoral Fellow at MIT ChemE)* | Indian Institute of Technology, Kharagpur, India (2025 – Ongoing)
  Formulated a 2D axisymmetric perturbative model under lubrication approximation to analyze non-linear solutal transport dynamics during hydrogel swelling.

Published Research Projects

- **Enhancing Electrokinetic Energy Harvesting via Rheology and Slip-Responsive Zeta Potential** (Dec '24 – Dec '25)
  Guide – Prof. Antarip Poddar, Department of Mechanical Engineering, IIT Dhanbad, India
  - Developed a theoretical framework investigating interactions between slip-responsive zeta potential and complex non-Newtonian rheology inside a microchannel.
- **PINNs for Flow Dynamics and Heat Transfer in Wavy Channels** (Jul '24 – Apr '25)
  Guide – Prof. Ameeya Kumar Nayak, Department of Mathematics, IIT Roorkee, India
  - Proposed a mixed variable deep-learning architecture reducing derivative orders in momentum equations for efficient modeling of flow field and validated against CFD; evaluated local Nusselt number enhancement.
- **Ammonia as a Hydrogen Carrier for Energy-Efficient Power Systems** (May '23 – Jul '24)
  Guide – Prof. Sandipan K. Das, Department of Mechanical Engineering, IIT Dhanbad, India
  - Analyzed ammonia decomposition pathways to produce hydrogen for steam turbine power generation using chemical looping technology, optimizing reactor parameters and flow rates.

Technical Skills

- **Programming Languages:** MATLAB, Fortran95, Python, C++
- **Simulation Tools & Software:** COMSOL Multiphysics, ANSYS Fluent, SolidWorks

Teaching, Mentoring and Other Experiences

- **Teaching Assistant**, Department of Mechanical Engineering, IIT Bombay (Jul '25 – Present)
  - **ME 306: Applied Thermodynamics** (Autumn 2026–2027)
    Conducted tutorial sessions, managed course logistics, held weekly office hours, and assisted with examination evaluations. *(Instructor: Prof. Sreedhara Sheshadri)*